Retail Customer Retention Analytics – WALMART

Problem Statement: At Walmart with increasing competition from Amazon, Target, and other retailers, retaining customers has become a critical challenge. Walmart collects vast data on customer purchases, loyalty programs, and online interactions, but their current reporting lacks analytical depth.

Objective: Develop an interactive Customer Retention Analytics Dashboard in Power BI using Walmart data that will:

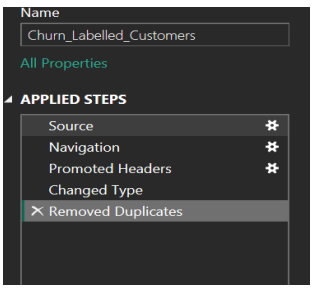
- Consolidate customer demographics, purchase history, store/e-commerce performance, and membership data.
- Enable dynamic segmentation of high value, repeat, and churned customers.
- Provide actionable insights to improve retention, loyalty engagement, and regional strategies

Dataset Description:

1. Customer_Demographics Columns: Customer_ID, Age, Gender, Region, Income_Level, Membership_Since, Preferred_Channel (Store/Online)
2. Customer_Transactions Columns: Transaction_ID, Customer_ID, Store_ID, Product_Category (Groceries, Electronics, Apparel, Home & Living), Transaction_Date, Amount, Promotion_Applied (Yes/No)
3. Store_Locations Columns: Store_ID, Store_Type (Supercenter, Neighborhood Market, Sam's Club, Online), Region, Opening_Year
4. Loyalty_Program Columns: Customer_ID, Loyalty_Tier (Basic, Plus, Premium, Elite), Points_Earned, Points_Redeemed
5. Churn_Labelled_Customers Columns: Customer_ID, Last_Purchase_Date, Churn_Flag (0 = Active, 1 = Churned), Churn_Reason (Inactivity, Competitor, Low Engagement)

Data Modelling and Cleaning: Before analysis, the data was cleaned and standardized.

- Handled duplicates, missing values, and ensured correct data types for each column in Power Query.



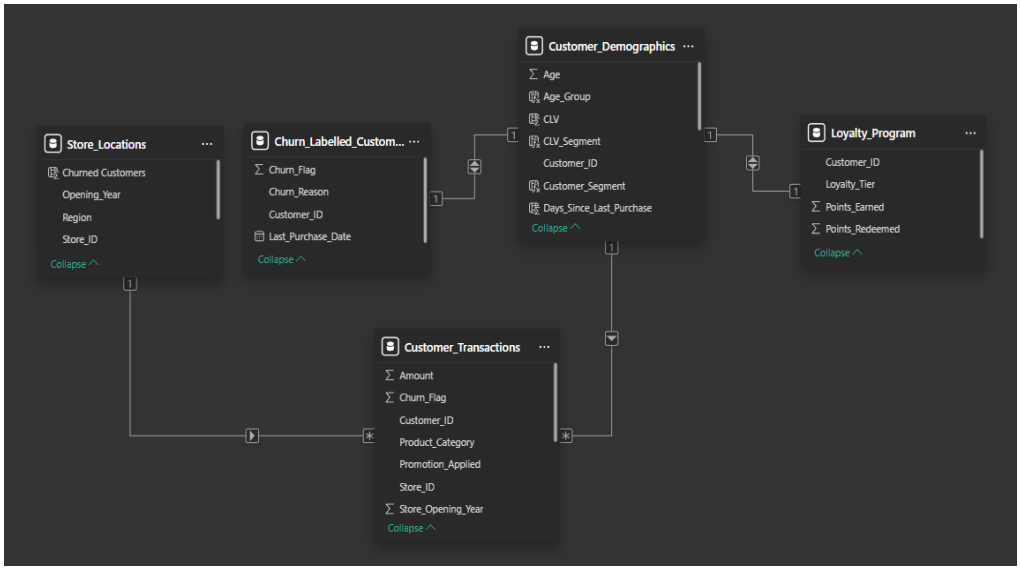
- Created calculated columns: Membership_Duration, Transaction_Year, Transaction_Month.

```
Membership_Duration = DATEDIFF([Membership_Since], TODAY(), YEAR)
```

```
Transaction_Year = YEAR(Customer_Transactions[Transaction_Date])
```

```
Transaction_Month = MONTH(Customer_Transactions[Transaction_Date])
```

- Created a basic Data Model view - One-to-Many: Customer_Demographics → Transactions, Loyalty_Program, Churn_Labelled_Customers and Many-to-One: Transactions → Store_Locations.



Churn & Retention Metrics:

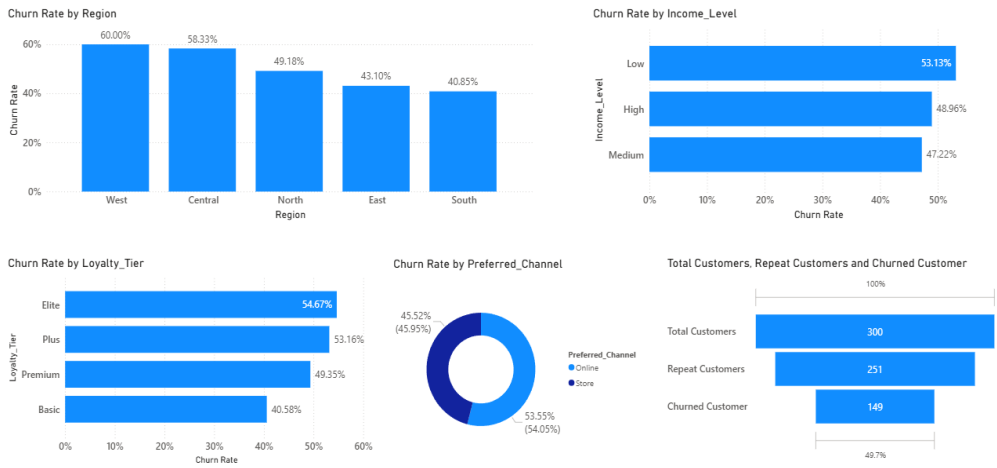
- Created Churn Rate KPI = (Churned Customers / Total Customers) * 100. Overall, 49.7% Churn Rate

```
Churned Customer = CALCULATE(COUNTROWS(Churn_Labelled_Customers), Churn_Labelled_Customers[Churn_Flag]=1)
```

```
Total Customers = COUNTROWS(Customer_Demographics)
```

```
Churn Rate = DIVIDE([Churned Customer],[Total Customers],0)
```

- Visualized Churn Rate by: Region, Income Group, Channel (Store/Online), Loyalty Tier
- Funnel Chart: Total Customers → Repeat Customers → Churned. It was found 300 : 251: 149



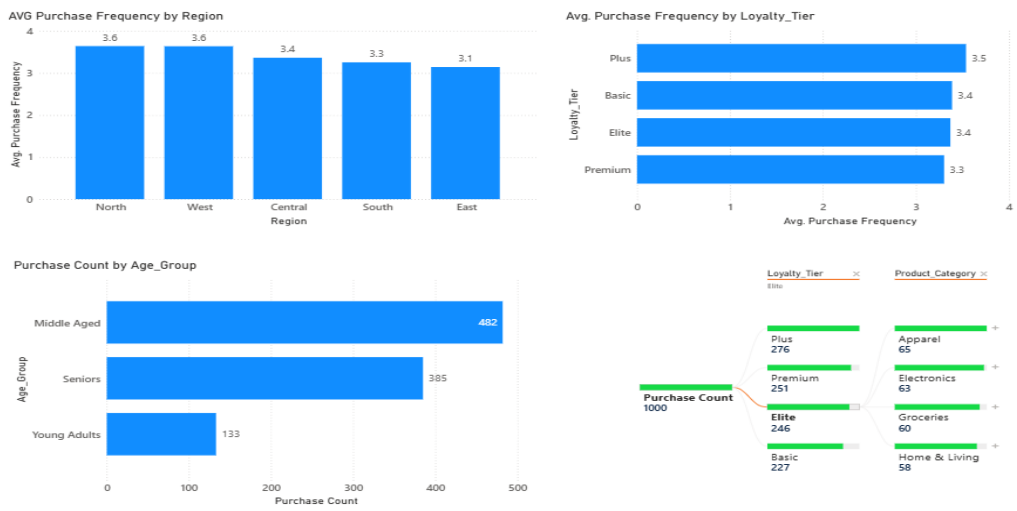
Repeat Purchase Analysis:

- Segmented customers: As Low-Tier for 0–3 purchases, Mid-Tier for 4–8 purchases & High-Tier for 9 and more purchases.

```
Purchase Count = CALCULATE( COUNT(Customer_Transactions[Transaction_ID]))
Customer_Segment = IF([Purchase Count]<4, "Low-Tier", IF([Purchase Count]<9,"Mid-Tier", "High-Tier"))
```

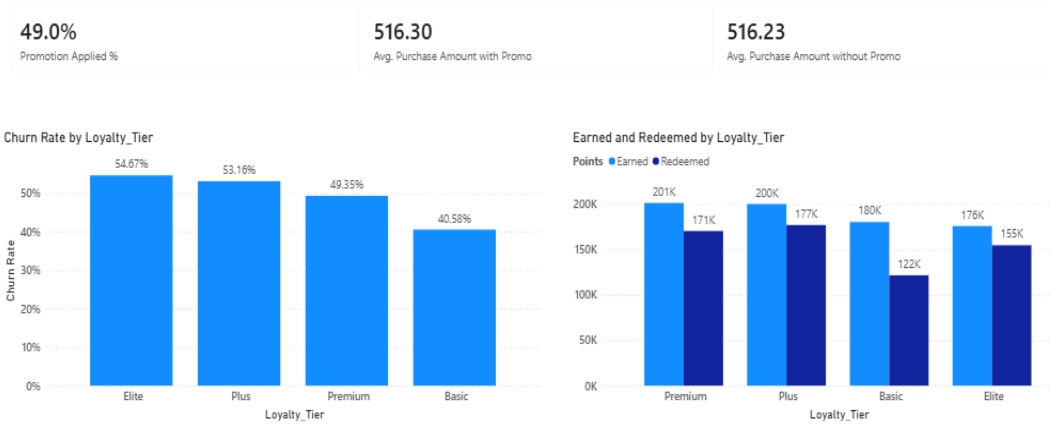
- Compared avg. purchase frequency by Region, Age Group, Loyalty Tier. Overall, 3.4 was the Average Purchase Frequency.
- Identified most purchased product categories by loyal customers: For the Premium, Plus and Basic Loyalty Tier it was Groceries (product category) and for the Elite Tier it was Apparel.

```
No_Of_Purchases = CALCULATE(COUNT(Customer_Transactions[Transaction_ID]))
Avg. Purchase Frequency = AVERAGE(Customer_Demographics[No_Of_Purchases])
```



Promotion & Loyalty Impact:

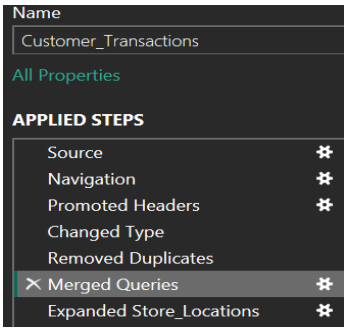
- Percentage of transactions with promotion applied was found to be 49%.
- Compared avg. purchase amount with promotions (516.30) vs without promotions (516.23)
- Visualized Churn rate across loyalty tiers, Points Earned vs Redeemed by Tier



- Recommendations to improve redemption & retention: -
  Effectiveness of the Promotion Applied can be increased as the avg. purchase amount does not differ much from the purchases with no promotions applied.
  Churn Rate is high for the Elite and Plus Tier customers. The churn reasons should be analysed carefully and so that high churning can be avoided in the future. If Inactivity is the foremost reason, then the customers inactive for a particular period should be notified or given some discount or limited time offer to make them active users again.

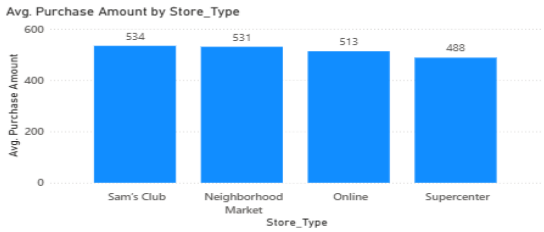
Store & Channel Performance vs Retention:

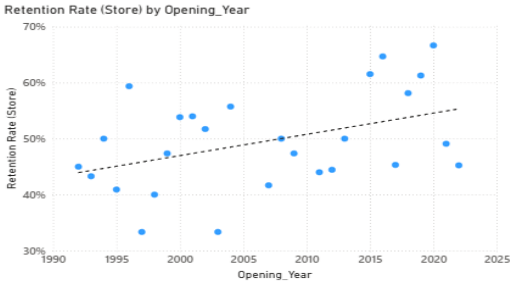
- Merged store data with transactions using merge queries in the power query editor.

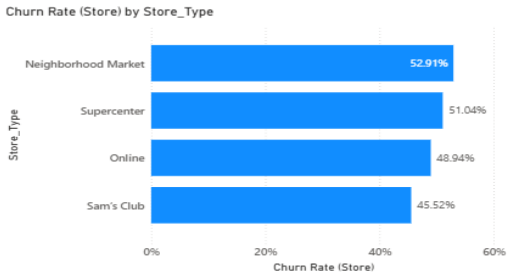


- Visualized Avg. transaction amount by Store Type, Churn rate by store type
- Correlation between store opening year & retention rate – It gives positive correlation and shows the latest stores hold better retention.

```
Avg. Purchase Amount = AVERAGE(Customer_Transactions[Amount])
```







Customer Lifetime Value (CLV) Analysis:

- CLV = Total Amount Spent / Membership Duration (Years). Overall, it was found to be 305.5 per year.

```
Total Amount = SUM(Customer_Transactions[Amount])
```

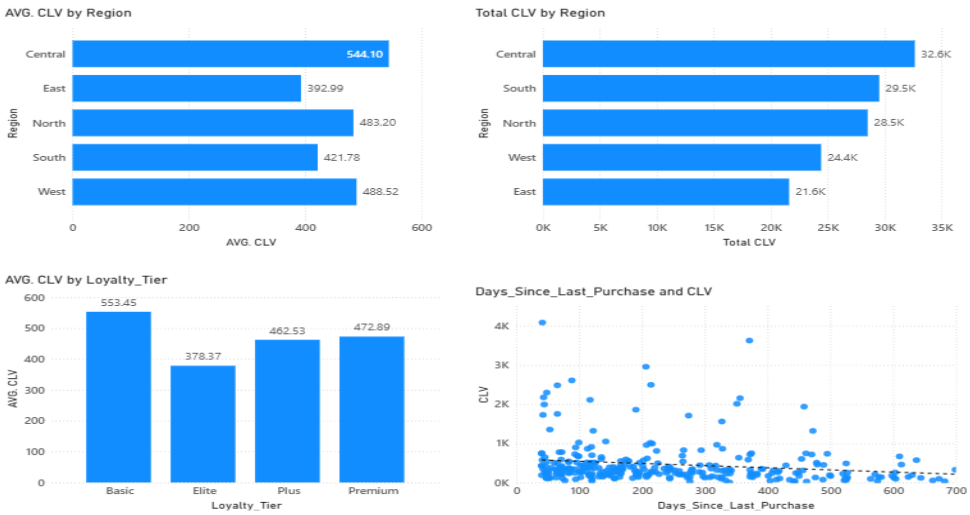
```
Membership Duration = SUM(Customer_Demographics[Membership_Duration])
```

```
CLV = DIVIDE([Total Amount],[Membership Duration],0)
```

- Segmented customers into Low, High CLV. Above Average CLV as High and below Average CLV value as Low.

```
CLV_Segment = IF([CLV]>AVERAGE(Customer_Demographics[CLV]),"High CLV", "Low CLV")
```

- Visualized CLV vs Days Since Last Purchase – A negative correlation was found out which means the lesser the days since last purchase higher the CLV.
- Visualized CLV by Loyalty Tier (Basic tier has the highest average CLV) and CLV by Region (Central has the highest CLV)



Final Dashboard:

Multi-page Power BI Report with Slicers:

- Page 1: KPIs (Churn, CLV, Repeat Rate)

- Page 2: Loyalty & Promotion Impact

- Page 3: Store/Channel Insights

- Page 4: Segmentation (Churned, Repeat, High-Value)

Top 3 recommendations for Walmart:

- Retention efforts should be focused on high CLV customers showing early inactivity signals, as they contribute more to revenue and are cheaper to retain than acquiring new customers. For this Walmart can launch **re-engagement campaigns**: Give personalized offers based on past categories. “We miss you” campaigns after inactivity threshold (like no purchase in 30–60 days then send targeted incentives).
- The customers who have Online as preferred channel are 17.6% more likely to churn than who prefer Stores. For this issue Walmart can look to improve UX and delivery speed, reduce cart abandonment through notifications and personalize suggestions to buy products. The Store type – Neighbourhood Market and Supercenter have more than 50% churn rate, these can be improved by better in-store experience, offer store-only promotions and optimizing inventory (avoid stockouts) while high-performing Stores like Sam’s Club (Highest average purchase amount and least churn rate) should be further leveraged for customer acquisition.
- The current loyalty program lacks structured behavioural rewards (based on frequent purchases, average order value, online engagement etc.), as promotions do not significantly influence spending by customers. A shift towards personalized, tier-based engagement strategies can improve both retention and customer lifetime value. For example, giving exclusive perks for Elite tier - early access to deals, free delivery, quick support.

Video Link - https://drive.google.com/file/d/1s8DpEO-JtRPlwyPUff5S2GtBxoJDz3t8/view?usp=sharing